

13-08-26 lab Assignment - 1

Q1) Task-1

using Arithmetic mean method
machine - A

Rewards: 3, 2, 4

$$Q(A) = (3 + 2 + 4) / 3$$

$$Q(A) = 9 / 3$$

$$Q(A) = 3.0$$

machine - B

Rewards: 6, 7

$$Q(B) = (6 + 7) / 2$$

$$Q(B) = 13 / 2$$

$$Q(B) = 6.5$$

machine - C

Rewards: 5, 4, 6

$$Q(C) = (5 + 4 + 6) / 3$$

$$Q(C) = 15 / 3$$

$$Q(C) = 5.0$$

$$Q(A) = 3.0$$

$$Q(B) = 6.5$$

$$Q(C) = 5.0$$

Task-2

Q2) Incremental update method

$$\text{formula} - Q_{\text{new}} = Q_{\text{old}} + (R - Q_{\text{old}}) / N$$

$$\text{initially: } Q(A) = 0, N(A) = 0$$

$$Q(B) = 0, N(B) = 0$$

$$Q(C) = 0, N(C) = 0$$

machine A

Rewards: 3, 2, 4

→ first reward = 3

$$N=1$$

$$Q(A) = 0 + (3-0)/1$$

$$Q(A) = 3$$

→ second reward = 2

$$N=2$$

$$Q(A) = 3 + (2-3)/2$$

$$Q(A) = 3 - 0.5$$

$$Q(A) = 2.5$$

→ third reward = 4

$$N=4$$

$$Q(A) = 2.5 + (4-2.5)/3$$

$$Q(A) = 2.5 + 0.5$$

$$Q(A) = 3.0$$

machine B

Rewards: 6, 7

→ first reward = 6

$$N=1$$

$$Q(B) = 0 + (6-0)/1$$

$$Q(B) = 6$$

→ second reward = 7

$$N=2$$

$$Q(B) = 6 + (7-6)/2$$

$$Q(B) = 6 + 0.5$$

$$Q(B) = 6.5$$

machine C

Rewards : 5, 4, 6

⇒ first reward = 5

$$N=1$$

$$Q(C) = 0 + (5 - 0) / 1$$

$$Q(C) = 5$$

⇒ second reward = 4

$$N=2$$

$$Q(C) = 5 + (4 - 5) / 2$$

$$= 5 - 0.5$$

$$= 4.5$$

⇒ third reward = 6

$$N=3$$

$$Q(C) = 4.5 + (6 - 4.5) / 3$$

$$= 4.5 + 0.5$$

$$Q(C) = 5.0$$

$$Q(A) = 3.0$$

$$Q(B) = 6.5$$

$$Q(C) = 5.0$$

Greedy selected machine = B